

Student Attendance System

The Problem Statement

- Manual attendance registers are extremely time-consuming and highly prone to human errors.
- Educational institutions struggle with inefficient, paper-based record management.
- Calculating attendance percentages manually is tedious and delays crucial administrative reporting.
- Tracking students with low attendance is difficult without an automated warning system.



System Objectives



Mark & Update

Allow teachers to seamlessly mark and update daily attendance records with minimal typing.



Auto Calculate

Automatically calculate attendance percentages with high precision to easily identify low attendance.



Secure Storage

Store student and attendance data persistently using a local database and retrieve it instantly when needed.

Key Technical Features

- SQLite Integration:** Utilizes persistent local storage with cascading deletes and strong referential integrity.
- Exception Handling:** Robust input validation traps errors (e.g., date formats, duplicate roll numbers).
- Batch Processing:** Uses REPLACE INTO to handle updates efficiently; automatically defaults unmarked students to 'Present'.
- CSV Export:** Seamlessly exports generated overall statistical reports into standard CSV formats.



System Workflow



1. Setup

Initialize the database and register students by assigning their Name, Roll Number, Class, and Subject.



2. Execution

Select class/subject and input only the absent students. The system automatically marks the remainder as present.

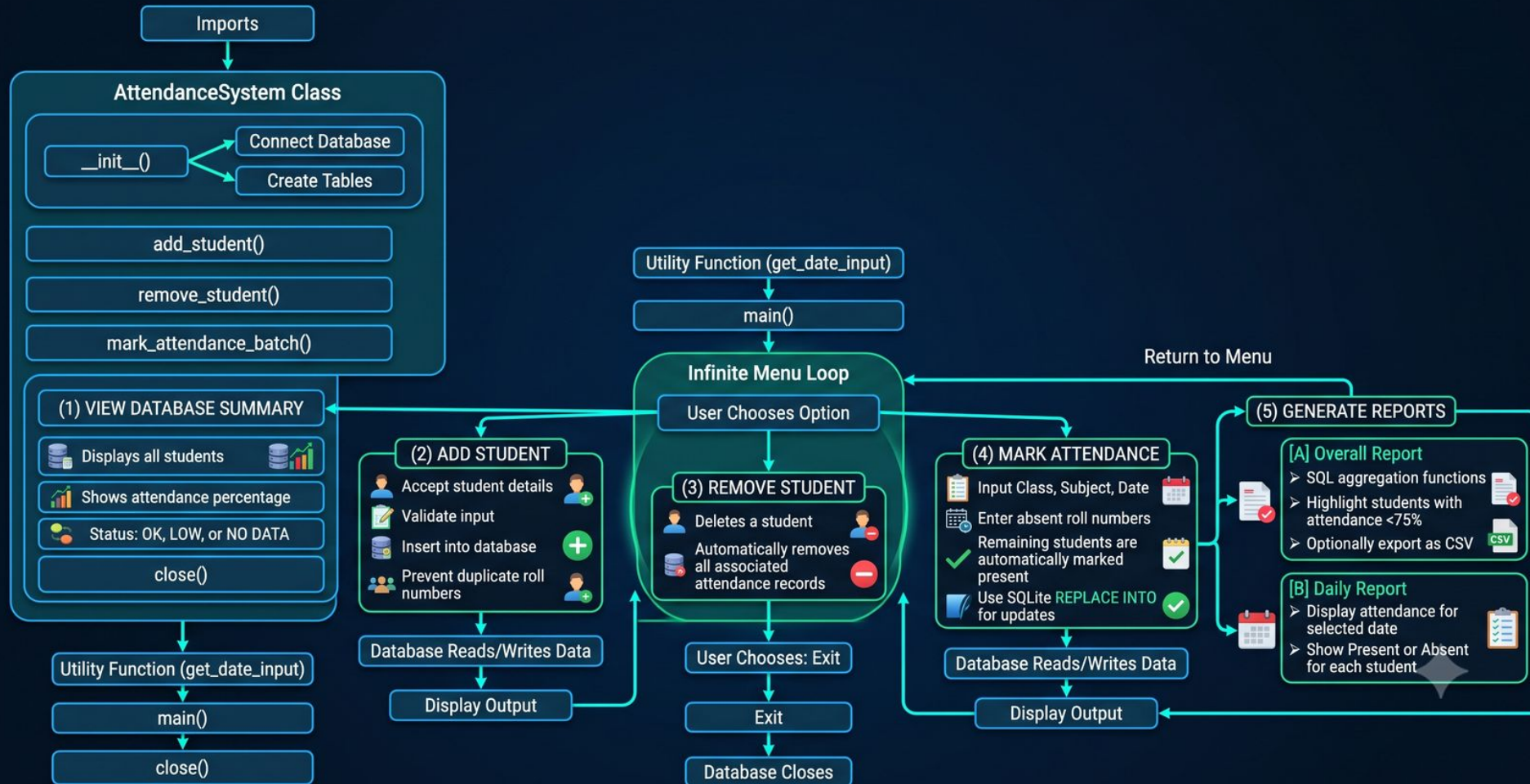


3. Analysis

Generate comprehensive daily reports or overall attendance percentage analytics in the terminal.

System Architecture

ATTENDANCE SYSTEM WORKFLOW DIAGRAM



Database Architecture

</> Students Table: Utilizes a composite primary key consisting of (roll_no, class_name, subject).

</> Attendance Table: Linked directly to students via Foreign Keys with ON UPDATE CASCADE ON DELETE CASCADE

</> This architecture ensures that if a student is dropped from a class, their entire attendance history is automatically purged

</> PRAGMA foreign_keys = ON is activated to enforce strict relational integrity.

Core Functions Overview

- `</>` **add_student()**: Safely inserts records, preventing app crashes on duplicates by catching the `sqlite3.IntegrityError`.
- `</>` **mark_attendance_batch()**: Optimizes input operations by only asking for absent roll numbers, and executes UPSERT logic.
- `</>` **generate_overall_report()**: Uses a LEFT JOIN alongside SQL aggregations to instantly compute historical percentages.
- `</>` **get_date_input()**: A strict validation loop ensuring all dates accurately adhere to the ISO-8601 subset (YYYY-MM-DD).

Reporting Capabilities

- Overall Report:** Aggregates historical data directly inside the SQLite engine for high efficiency and speed.
- Daily Report:** Uses an INNER JOIN to fetch exact attendance status matches for a targeted specific date.
- Database Summary:** Provides a global view of all classes, actively flagging students with less than 75% attendance.
- Export Option:** Uses safe file I/O to generate clean CSV files for external administrative use.



Project Impact

Administrative Efficiency

The system significantly reduces the paperwork and time required to manage daily attendance. The automated calculations prevent human errors and speed up the grading and reporting processes.

Enhanced User Experience

It provides a fast, simple, and intuitive Command-Line Interface (CLI). Educators can quickly adopt the system with minimal training, drastically improving their daily administrative workflow.

Thank

Are there any questions about the system?

You!

TEAM

MEMBERS • ANASWINI

• GRESHMA

• BASHITHA

• RICHA

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• PRAKHAR